

View Reviews

Paper ID

634

Paper Title

Characterizing Speculative Decoding Dynamics for Large Language Models on Consumer-Class GPUs

Reviewer #1

Questions**2. Comments for the author:**

Strengths: I appreciate the experimental design. It is well planned, clear, and well controlled. The paper provides useful system-level insights, especially that larger "k" does not improve performance on this hardware and that deterministic decoding is usually faster than stochastic decoding. The writing is ambitious and the paper gives a coherent framework for discussing latency, acceptance, quality drift, and tail behavior together. The most interesting empirical finding is that increasing "k" from 4 to 8 or 16 reduces speedup on this hardware, even though the accepted block size increases. This is useful deployment guidance and shows that higher acceptance or larger blocks do not automatically translate into better wall-clock performance.

Weaknesses: The main weakness is that the proposed adaptive method does not outperform the best simple fixed setting. The paper itself shows that a basic fixed configuration remains the fastest option, so the new controller reads more like a robustness mechanism than a true performance improvement. In addition, the evaluation is limited to one GPU, the RTX 4090, and one model family, Qwen2.5, so it is unclear whether the same trends would hold on other GPUs such as the RTX 3090, RTX 4080, or A100, or with other model families such as Llama or Mistral. I was also unconvinced by the CUDA graph discussion. The paper presents a CUDA-graph-capable setup, but later states that the capture rate was zero, which means the main results did not actually benefit from CUDA graph execution.

Reviewer #2

Questions**2. Comments for the author:**

This paper presents a careful and practically motivated empirical characterization of speculative decoding on consumer-class GPUs, focusing on RTX4090 deployment scenarios with Qwen2.5-family models. The work is valuable because much of the

existing speculative decoding literature emphasizes datacenter-scale systems or introduces new architectures, whereas this paper studies realistic operating points for practitioners running local inference workloads.

The strongest contribution is the controlled experimental methodology. The use of frozen manifests, paired comparisons, regime-matched baselines, bootstrap confidence intervals, and explicit reporting of acceptance/block-size dynamics gives the study a level of rigor often missing from systems benchmarking papers. The analysis of the non-monotonic relationship between acceptance efficiency and end-to-end speedup is particularly useful and well explained. The observation that larger k increases accepted block size while still hurting throughput on RTX4090-class hardware is insightful and deployment-relevant.

The paper is also commendably transparent about limitations. In particular, explicitly reporting that CUDA graph capture effectively failed in the main run family and that the observed gains come from fallback execution improves credibility. The discussion framing ACSD as a “tail-latency guardrail” rather than a throughput-maximizing algorithm is appropriately modest and technically consistent with the results.

That said, the paper has several weaknesses that prevent it from being a stronger accept.

First, the novelty is somewhat limited. Most of the technical core is an empirical benchmarking study of existing speculative decoding techniques rather than a fundamentally new algorithmic contribution. ACSD is presented more as an engineering policy/controller than a new decoding method, and its gains over the best fixed policy are not demonstrated. The paper honestly acknowledges this, but it still weakens the contribution relative to top-tier systems or ML venues.

Second, the scope is narrow. All experiments are conducted on a single GPU type (RTX4090) and one model family (Qwen2.5). Because speculative decoding behavior is highly hardware- and architecture-dependent, it is difficult to determine how generalizable the findings are. Even adding one additional hardware class (e.g., RTX 3090, A100, or AMD GPU) would substantially strengthen the claims.

Third, some aspects of the evaluation could be improved:

The paper lacks variance across repeated independent runs/seeds beyond bootstrap resampling over samples.

TTFT and TPOT are defined but not deeply analyzed despite being important deployment metrics.

The quality evaluation is somewhat weak for summarization; relying only on ROUGE-L is limiting.

Statistical testing methodology is only lightly specified (e.g., exact paired tests, multiple-comparison corrections, effect sizes).

Fourth, parts of the ACSD section read more like a position/design paper than a fully validated systems contribution. The FSM formalization is reasonable, but many policy thresholds and heuristics appear manually chosen, and there is little ablation analysis on rescue thresholds, switching policies, or cooling windows.

There are also several presentation issues:

The paper is occasionally repetitive, especially in Sections VII–VIII.

Some claims are overstated relative to the experimental breadth (e.g., broader deployment implications).

Minor grammatical issues appear throughout (e.g., “How much end-to-end speedup does vanilla speculative decoding delivers...”).

Figure 2 is information-dense and somewhat difficult to interpret in grayscale print form.

Overall, however, the paper is technically sound, reproducible, practically relevant, and well organized. While the novelty ceiling is limited, the work provides useful deployment guidance and careful empirical evidence for consumer-GPU speculative decoding behavior.

Reviewer #3

Questions

2. Comments for the author:

The paper presents a controlled study to characterize speculative decoding dynamics for Large Language Models on consumer-class GPUs. While it is clear that the results from the paper has a great practical contribution as a deployment-oriented guidance, the novelty and theoretical contribution of the work is not clear. The authors should consider revising the paper to also highlight the theoretical part of the work. In addition, it would benefit the work if other methods can also be mentioned and compared along the proposed method used for the study. Moreover, the format in the abstract should be revised, every word in the abstract should be bolded, and there should be a brief introduction for related work so that it would be easier for the readers to follow.